

PRADYUMNA UPADHYAY

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EDUCATION

University of Maryland, College Park, MD | *Master of Engineering in Software Engineering*
Devi Ahilya Vishwavidyalaya, Indore, India | *Bachelor of Computer Engineering*

Aug 2023 - May 2025
Jul 2017 - Jun 2021

SKILLS

- **Programming Languages and Fundamentals:** C++, C#, JavaScript, Python, Typescript, Data Structures
- **Web and API Development:** Angular, Flask, FastAPI, HTML/CSS, jQuery, Next.js, Node.js, OAuth, React, Redux, Rest API
- **Cloud and DevOps Tools:** Amazon Web Services (EC2, EMR, Cloud9, S3), Git, GitHub, Postman, Visual Studio Code
- **Databases and ORM:** MySQL, MSSQL, MongoDB, SQLAlchemy
- **Data Processing and Analysis:** ArcGIS, Geopandas, Geopy, Folium, OpenStreetMap, Pandas, PySpark, Esri Experience Builder
- **AI and APIs:** Prompt Engineering, LLM Integration, NLP, API Integration, API Orchestration, JSON Handling, Asynchronous Programming
- **Project Management Tools:** Agile, Confluence, JIRA, Slack

WORK EXPERIENCE

University of Maryland | Software Engineer Intern

Oct 2023 - Present

- Led two major projects improving pedestrian safety and crash data analysis, showcasing expertise in **geospatial data processing and fullstack web application development**.
- **Reduced data interpretation errors by 25%** through an interactive Folium/OpenStreetMap visualization for high-risk pedestrian areas.
- Optimized data processing pipeline using PySpark, and AWS EMR, **cutting processing time by 30%** for **20** terabytes of geospatial data.
- Built a **React and Flask-based web application** analyzing **10,000+** MD vehicle crashes, combining Esri's Experience Builder for enhanced visualization and **increasing data accessibility by 20%** through PDF/Excel exports.
- Implemented CI/CD pipeline on Azure, **improving deployment efficiency and reliability**.
- Created a custom file storage, **cutting down input redundancy by 50%** and enhanced user experience by allowing saved analysis sessions.
- Demonstrated **product development** skills by **optimizing user experience**, troubleshooting complex issues, and refining data visualization for stakeholders.

LTIMindtree | Software Engineer

Aug 2021 - Jul 2023

- Spearheaded **enhancements for a ASP.Net MVC-based pricing web tool**, live across **7** countries and **200+** user base, restructuring SQL database tables to **condense size by 15%** and decreasing data uploading time by **85%**.
- Architected back-end front-end integration, providing technical support and serving as sole point of contact between Development, Testing, and DevOps teams, fostering cross-team collaboration, hence, **increasing application performance by 30%** and streamlining issue resolution.
- Optimized software development life cycle by collaborating with cross-functional teams, **reducing bug incidents by 25%**, expediting release cycles by **30%**, and improving system stability through problem-solving.
- Exemplified professional mentoring by on boarding and guiding **4** new peers by conducting knowledge sessions to train them on tools and technologies operated within and outside project scope.

PROJECTS

Travel Buddy

Bitcamp 2025

- Built a **full-stack AI-driven travel assistant in under 36 hours**, delivering real-time personalized itineraries and contextual travel insights.
- **Shaped prompt workflows, synchronized external APIs, and fine-tuned UI/UX** to ensure cohesive end-to-end functionality.
- Architected a **FastAPI backend incorporating Gemini 2.0 Flash for extracting structured trip data and orchestrated third-party API api services** for weather, flights, dining, and attractions.
- Built a responsive interface using React and Tailwind CSS, refined layout for smoother user interaction.

UpSkilled - Learning Management System

Aug 2024 - Dec 2024

- Constructed and deployed a comprehensive **Learning Management System (LMS)** utilizing React, Redux, and Spring (Java). The platform features tailored dashboards and functionalities for instructors, employees, and administrators.
- Introduced interactive instructor features to streamline course and assignment management. By integrating embedded file viewers for PDFs and other submissions, **grading efficiency improved by 40%**.
- **Optimized employee learning experience** by creating intuitive dashboards. These dashboards allow employees to monitor their progress, access enrolled courses, and view detailed assignment information. As a result, course completion rates increased by **25%**.
- **Implemented responsive design** for cross-device compatibility, boosting mobile platform usage by **35%**.

SmartCam - Facial Recognition Attendance System

Aug 2024 - Dec 2024

- Co-led development of a dual-application **attendance tracking system**, including a desktop application with Python and OpenCV, achieving **95% facial recognition accuracy** in diverse lighting conditions and identifying up to **10 faces per frame**.
- Web application: Developed a **web platform with React and Node** to create user-friendly dashboards for faculty and admin, enabling personalized access to attendance data and management features.
- Partnered in **integrating MongoDB Atlas** for seamless data retrieval and storage, supporting comprehensive analysis and reporting.
- Work closely with a teammate, ensuring **cohesive project execution and effective deployment**.